Exposé II. Existence of global resolutions

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1. General globalization criteria

1.0

In this section we are given a site S, a flat abelian S-category $\mathcal{C}$ (I 1.1), and a strictly full additive sub-S-category $\mathcal{C}_0$ (I 1.1). We assume that $\mathcal{C}_0$ is locally quasi-liftable in $\mathcal{C}$ (I 1.2) and locally quasi-stable under kernels of epimorphisms (I 1.2).

Proposition 1.1. Let $S \in \mathrm{ob}(S)$, and suppose that:

(i) $\mathcal{C}_{0S}$ is quasi-liftable in $\mathcal{C}_S$ (I 1.2), where $\mathcal{C}_S$ is regarded as a category fibered over a punctual site;

(ii) every object of $\mathcal{C}_S$ which is of finite $\mathcal{C}_0$-type (I 1.2), respectively of finite $\mathcal{C}_0$-presentation (I 2.8), is of finite $\mathcal{C}_{0S}$-type.

Then:

a) $\mathcal{C}_{0S}$ is quasi-stable under kernels of epimorphisms (I 1.2), where $\mathcal{C}_{0S}$ is regarded as a category fibered over a punctual site.

b) Let $E \in \mathrm{ob}\,\mathrm{D}^-(\mathcal{C}_S)$ and $n \in \mathbb{Z}$. If E is n-$\mathcal{C}_0$-pseudo-coherent (I 2.2), then E is n-$\mathcal{C}_{0S}$-pseudo-coherent, respectively $(n+1)$-$\mathcal{C}_{0S}$-pseudo-coherent. In other words,

$$\mathrm{D}^-(\mathcal{C}_S)_{n-\mathcal{C}_0\text{-coh}} = \mathrm{D}^-(\mathcal{C}_S)_{n-\mathcal{C}_{0S}\text{-coh}},$$

and, respectively,

$$\mathrm{D}^-(\mathcal{C}_S)_{n-\mathcal{C}_0\text{-coh}} \subset \mathrm{D}^-(\mathcal{C}_S)_{(n+1)-\mathcal{C}_{0S}\text{-coh}} \subset \mathrm{D}^-(\mathcal{C}_S)_{(n+1)-\mathcal{C}_0\text{-coh}},$$

inclusions of strictly full subcategories of $\mathrm{D}(\mathcal{C}_S)$, the inclusion on the right being trivial.

c) For an object $E \in \mathrm{ob}\,\mathrm{D}(\mathcal{C}_S)$ to be pseudo-coherent relative to $\mathcal{C}_0$ (I 2.3), it is necessary and sufficient that it be pseudo-coherent relative to $\mathcal{C}_{0S}$; equivalently,

$$\mathrm{D}^-(\mathcal{C}_S)_{\mathcal{C}_0\text{-coh}} = \mathrm{D}^-(\mathcal{C}_S)_{\mathcal{C}_{0S}\text{-coh}}.$$

Moreover, the canonical functor

$$\mathrm{K}^-(\mathcal{C}_{0S})/\mathrm{K}^{-,\delta}(\mathcal{C}_{0S}) \longrightarrow \mathrm{D}(\mathcal{C}_S),$$

where $\mathrm{K}^{-,\delta}(\mathcal{C}_{0S})$ denotes the thick subcategory of $\mathrm{K}^-(\mathcal{C}_{0S})$ formed by acyclic complexes, is fully faithful and has as essential image $\mathrm{D}^-(\mathcal{C}_S)_{\mathcal{C}_0\text{-coh}}$. If $\mathcal{C}_{0S}$ is liftable in $\mathcal{C}_S$ (I 1.2), that is, if the objects of $\mathcal{C}_{0S}$ are projective, then $\mathrm{K}^{-,\delta}(\mathcal{C}_{0S})$ is reduced to zero objects, and consequently the functor

$$\mathrm{K}^-(\mathcal{C}_{0S}) \longrightarrow \mathrm{D}(\mathcal{C}_S)$$

is fully faithful and has essential image $\mathrm{D}^-(\mathcal{C}_S)_{\mathcal{C}_0\text{-coh}}$.

Proof. a) Let

$$0 \longrightarrow L \longrightarrow L^0 \longrightarrow \cdots \longrightarrow L^m \longrightarrow 0$$

be an exact sequence in $\mathcal{C}_S$, with $L^i \in \operatorname{ob} \mathcal{C}_{0S}$ for all i. It is enough to prove that L is of finite $\mathcal{C}_{0S}$-type. By (I 2.2), L is $\mathcal{C}_0$-pseudo-coherent, hence, by (I 2.9), of finite $\mathcal{C}_0$-presentation; the assertion follows from (ii).

b) We prove by descending induction on i that E is i-$\mathcal{C}_{0S}$-pseudo-coherent for $i \geq n$, respectively for $i \geq n+1$. One may already suppose that E is strictly $(n+1)$-$\mathcal{C}_{0S}$-pseudo-coherent, respectively strictly $(n+2)$-$\mathcal{C}_{0S}$-pseudo-coherent (I 2.1); the point is then to prove that E is n-$\mathcal{C}_{0S}$-pseudo-coherent, respectively $(n+1)$-$\mathcal{C}_{0S}$-pseudo-coherent. Thus there is an exact sequence

$$0 \longrightarrow E' \longrightarrow E \longrightarrow E'' \longrightarrow 0,$$

where E' is strictly $\mathcal{C}_{0S}$-perfect (I 2.1) and $E''^i = 0$ for $i > n$, respectively for $i > n+1$. Replacing E by E'', one may therefore suppose (I 2.6) that $E^i = 0$ for $i > n$, respectively for $i > n+1$. Then, by (I 2.10 a)), $H^n(E)$, respectively $H^{n+1}(E)$, is of finite $\mathcal{C}_0$-type, respectively of finite $\mathcal{C}_0$-presentation. Hence, by (ii), $H^n(E)$, respectively $H^{n+1}(E)$, is of finite $\mathcal{C}_{0S}$-type. Applying (I 2.10 b)) again, one obtains that E is n-$\mathcal{C}_{0S}$-pseudo-coherent, respectively $(n+1)$-$\mathcal{C}_{0S}$-pseudo-coherent. The hypothesis (i) and part a) have been used implicitly in the references to (I 2.6 and 2.10).

c) The first assertion follows from b), the second from (I 2.7), and the last is evident.

Proposition 1.2. Suppose that $\mathcal{C}_0$ is locally liftable in $\mathcal{C}$ (I 1.2) and locally stable under kernels of epimorphisms (I 1.2), and make the hypotheses (1.1 (i) and (ii), respectively). Let $\mathcal{C}_{1S}$ denote the strictly full subcategory of $\mathcal{C}_S$ formed by the objects which are locally in $\mathcal{C}_0$. Then:

a) $\mathcal{C}_{1S}$ is stable under kernels of epimorphisms (I 1.2) and is quasi-liftable in $\mathcal{C}_S$ (I 1.2), as a category fibered over a punctual site.

b) Let $E \in \operatorname{ob} \mathrm{D}(\mathcal{C}_S)$ and let $[m, n]$ be an interval of $\mathbb{Z}$. For E to have $\mathcal{C}_0$-perfect amplitude contained in $[m, n]$, it is necessary and sufficient that E be isomorphic, in $\mathrm{D}(\mathcal{C}_S)$, to a complex F such that $F^i = 0$ for $i \notin [m, n]$, $F^i \in \operatorname{ob} \mathcal{C}_{0S}$ for $i > m$, and $F^m \in \operatorname{ob} \mathcal{C}_{1S}$.

c) The canonical functor
$$\mathrm{K}^b(\mathcal{C}_{1S})/\mathrm{K}^{b,\delta}(\mathcal{C}_{1S}) \longrightarrow \mathrm{D}(\mathcal{C}_S)$$
is fully faithful and has as essential image the full subcategory of $\mathrm{D}(\mathcal{C}_S)$ formed by complexes of finite $\mathcal{C}_0$-perfect amplitude.

d) If $\mathcal{C}_{1S}$ is not merely quasi-liftable but liftable in $\mathcal{C}_S$, one may replace, in b), the expression "E is isomorphic to F in $\mathrm{D}(\mathcal{C}_S)$" by "there exists a quasi-isomorphism $F \to E$"; and, in c), $\mathrm{K}^{b,\delta}(\mathcal{C}_{1S})$ is reduced to zero objects.

Proof. a) It is clear that $\mathcal{C}_{1S}$ is stable under kernels of epimorphisms. On the other hand, the objects of $\mathcal{C}_{1S}$ are $\mathcal{C}_0$-pseudo-coherent, hence $\mathcal{C}_{0S}$-pseudo-coherent by (1.1 c)); the second assertion follows from (I 2.14), taking S to be the punctual site.

b) It suffices to prove necessity. By (1.1 c)), E is already $\mathcal{C}_{0S}$-pseudo-coherent. Since E is acyclic in degree $> n$, one may, after replacing E by an isomorphic complex in $\mathrm{D}(\mathcal{C}_S)$, suppose (I 2.10) that E is strictly $(m+1)$-$\mathcal{C}_{0S}$-pseudo-coherent and that $E^i = 0$ for $i > n$. Since E is acyclic in degree $< m$, one may also replace E by its truncation

$$0 \longrightarrow E^m/B^m \longrightarrow E^{m+1} \longrightarrow \cdots \longrightarrow E^n \longrightarrow 0,$$

so that $E^i = 0$ for $i < m$. Then, by (I 4.16), E^m is locally an object of $\mathcal{C}_0$, and therefore an object of $\mathcal{C}_{1S}$.

c) In view of (1.2 a)), the assertion follows from (1.2 b)) and (I 4.12.2). d) The fact that $\mathrm{K}^{b,\delta}(\mathcal{C}_{1S})$ is reduced to zero objects is evident. The other assertion follows from (I 4.6), applied over the punctual site.

2. Application to certain categories of modules

2.0. Notation

Let (S, A) be a ringed topos. For $U \in \operatorname{ob} S$, write $\mathrm{Mod}(U)$ for the category of left A_U-modules, and $\mathrm{Lib}(U)$, respectively $\mathrm{Loclib}(U)$, for the full subcategory of $\mathrm{Mod}(U)$ formed by free A_U-modules, respectively modules locally direct factors of free modules, of finite type.

2.1. Lifting lemmas

Lemma 2.1.1 (cf. EGA 0_{I}, 5.2.3). Let (S, A) be a ringed topos, and let $S \in \operatorname{ob} S$. Suppose that S is quasi-compact, that is, every covering family $(S_i \to S)$ is dominated by a finite covering family.

a) Let F be a left A_S-module which is the filtered inductive limit, over a filtered category Λ, of A_S-modules F_λ, and let

$$u = \varinjlim u_\lambda : F \longrightarrow G$$

be an epimorphism, where G is an A_S-module of finite type. Then there exists $\lambda \in \Lambda$ such that u_λ is an epimorphism.

b) Let F be a left A_S-module generated by its sections, that is, such that there exists an epimorphism $A_S^{(I)} \to F$ for a suitable set of indices I. If F is of finite type, then there exists an epimorphism $A_S^p \to F$ for some $p \in \mathbb{N}$.

The proof of a) is analogous to the proof of the cited result and is left to the reader; b) follows immediately from a).

Lemma 2.1.2. Let (S, A) be a ringed topos, let $\mathcal{C}$ be the S-category fibered in left A-modules, let $\mathcal{C}_0$ be the fibered subcategory of free A-modules of finite type (I 1.3 d)), let $S \in \operatorname{ob} S$, and let $\mathcal{C}'_S$ be a strictly full additive subcategory of $\mathcal{C}_S$ containing $\mathcal{C}_{0S}$ and stable under kernels, cokernels, and extensions. Suppose that

$$H^1(S, F) = 0 \quad \text{for every } F \in \operatorname{ob} \mathcal{C}'_S.$$

Then:

a) Every exact sequence in $\mathcal{C}_S$

$$0 \longrightarrow E \longrightarrow F \longrightarrow G \longrightarrow 0,$$

such that $\mathrm{Hom}(G, E) \in \operatorname{ob} \mathcal{C}'_S$ and such that G is locally a direct factor of a free module of finite type, is split. The category $\mathcal{C}_{0S}$ is liftable in $\mathcal{C}'_S$ (I 2.2).

b) Let $\mathrm{Mod}(A(S))$ denote the category of $A(S)$-modules. The functor

$$\Gamma_S : \mathcal{C}'_S \longrightarrow \mathrm{Mod}(A(S))$$

is exact. Moreover:

(i) if every object F of $\mathcal{C}'_S$ admits a finite global presentation

$$A_S^q \longrightarrow A_S^p \longrightarrow F \longrightarrow 0,$$

then Γ_S is fully faithful and has as essential image the full subcategory of $\mathrm{Mod}(A(S))$ formed by $A(S)$-modules of finite presentation;

(ii) if $\mathcal{C}'_S$ contains the free A_S-modules, if every object F of $\mathcal{C}'_S$ admits a global presentation

$$A_S^{(J)} \longrightarrow A_S^{(I)} \longrightarrow F \longrightarrow 0,$$

and if Γ_S commutes with filtered inductive limits, then Γ_S is an equivalence of categories.

Proof. a) Since G is locally a direct factor of a free module of finite type, the spectral sequence

$$E_2^{pq} = H^p(S, \mathcal{E}xt^q(G,E)) \Longrightarrow \mathrm{Ext}^{p+q}(G,E)$$

degenerates (I 1.3.1) and gives an isomorphism

$$H^1(S, \mathrm{Hom}(G,E)) \simeq \mathrm{Ext}^1(G,E),$$

whence the first assertion. The second is a particular case, since for $G \in \mathrm{ob}\,\mathcal{C}_{0S}$ and $E \in \mathrm{ob}\,\mathcal{C}'_S$ one has $\mathrm{Hom}(G,E) \in \mathrm{ob}\,\mathcal{C}'_S$.

b) The first assertion is trivial. Let us prove, in case (i) or (ii), the full faithfulness of $\Gamma_S : \mathcal{C}'_S \to \mathrm{Mod}(A(S))$. Let $F, G \in \mathrm{ob}\,\mathcal{C}'_S$ and let

$$A_S^{(J)} \longrightarrow A_S^{(I)} \longrightarrow F \longrightarrow 0 \tag{$*$}$$

be a global presentation of F, with I and J finite in case (i). Applying Γ_S, which is exact on $\mathcal{C}'_S$ and, in case (ii), commutes with filtered inductive limits, one obtains an exact sequence

$$A(S)^{(J)} \longrightarrow A(S)^{(I)} \longrightarrow \Gamma_S(F) \longrightarrow 0. \tag{$**$}$$

Applying the left exact functors $\mathrm{Hom}_{\mathcal{C}_S}(-,G)$ and $\mathrm{Hom}_{A(S)}(-,\Gamma_S(G))$ to $(*)$ and $(**)$ gives a commutative diagram with exact rows:

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{Hom}_{\mathcal{C}_S}(F,G) \arrow[r] \arrow[d] & \mathrm{Hom}_{\mathcal{C}_S}(A_S^{(I)},G) \arrow[r] \arrow[d, "\sim"] & \mathrm{Hom}_{\mathcal{C}_S}(A_S^{(J)},G) \arrow[d, "\sim"] \\
0 \arrow[r] & \mathrm{Hom}_{A(S)}(\Gamma_S F, \Gamma_S G) \arrow[r] & \mathrm{Hom}_{A(S)}(A(S)^{(I)}, \Gamma_S G) \arrow[r] & \mathrm{Hom}_{A(S)}(A(S)^{(J)}, \Gamma_S G).
\end{tikzcd}$$

The two vertical arrows on the right are visibly isomorphisms, hence so is the arrow on the left. The description of the essential image in cases (i) and (ii) is immediate and is left to the reader.

Lemma 2.1.3. Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories, and let $\varphi : \mathcal{A} \to \mathcal{B}$ be an exact functor.

a) For $M \in \mathrm{ob}\,\mathcal{A}$, the following conditions are equivalent:

(i) for every $L \in \mathrm{ob}\,\mathcal{A}$, the canonical map induced by φ

$$\mathrm{Ext}^n_{\mathcal{A}}(L,M) \longrightarrow \mathrm{Ext}^n_{\mathcal{B}}(\varphi L, \varphi M)$$

is an isomorphism for all n;

(ii) for every $L \in \operatorname{ob} \mathcal{A}$, the canonical map

$$\operatorname{Hom}_{\mathcal{A}}(L, M) \longrightarrow \operatorname{Hom}_{\mathcal{B}}(\varphi L, \varphi M)$$

is an isomorphism, and for every $n \geq 1$ and every $x \in \operatorname{Ext}^n_{\mathcal{B}}(\varphi L, \varphi M)$ there exists an epimorphism $L' \to L$ in $\mathcal{A}$ which kills x.

b) The following conditions are equivalent:

(i) for all $L, M \in \operatorname{ob} \mathcal{A}$ and all $n \in \mathbb{Z}$, the canonical map

$$\operatorname{Ext}^n_{\mathcal{A}}(L, M) \longrightarrow \operatorname{Ext}^n_{\mathcal{B}}(\varphi L, \varphi M)$$

is an isomorphism;

(ii) the canonical functor induced by φ

$$\mathrm{D}^b(\mathcal{A}) \longrightarrow \mathrm{D}^b(\mathcal{B})$$

is fully faithful and has as essential image the full subcategory of $\mathrm{D}^b(\mathcal{B})$ formed by complexes F such that $H^i(F)$ is in the essential image of φ for every i.

Proof. For (i)⇒(ii) in a), it is enough to show that, if $n \geq 1$, $L \in \operatorname{ob} \mathcal{A}$, and $y \in \operatorname{Ext}^n_{\mathcal{A}}(L, M)$, then there exists an epimorphism $L' \to L$ killing y. Using Yoneda's interpretation ([18], III 3.2), y is the class of an exact sequence

$$0 \longrightarrow M \longrightarrow E_{n-1} \longrightarrow \cdots \longrightarrow E_0 \longrightarrow L \longrightarrow 0,$$

and the epimorphism $E_0 \to L$ kills y ([18], III 3.2.5 and 3.2.8). Conversely, put

$$F^n(L) = \operatorname{Ext}^n_{\mathcal{A}}(L, M), \qquad G^n(L) = \operatorname{Ext}^n_{\mathcal{B}}(\varphi L, \varphi M).$$

The effaceability hypothesis says, for $n \geq 1$ and $L \in \operatorname{ob} \mathcal{A}$, that

$$G^n(L) = \varinjlim \operatorname{Coker} igl(G^{n-1}(L_0) \to G^{n-1}(L_1)),$$

the limit being taken over the exact sequences $L_1 \to L_0 \to L \to 0$. As above, one has the same formula for $F^n(L)$. Since $F^0 \to G^0$ is an isomorphism, induction on n shows that the morphism $F^n \to G^n$ induced by φ is an isomorphism for every n. The equivalence in b) follows by an immediate dévissage, left to the reader.

Remarks 2.1.3.1. There is of course a dual version of a), which we leave to the reader to state. On the other hand, if $\mathcal{A}$ and $\mathcal{B}$ have enough injectives, the conditions (i) and (ii) of b) are equivalent to:

(iii) for $L \in \operatorname{ob} \mathrm{D}^-(\mathcal{A})$ and $M \in \operatorname{ob} \mathrm{D}^+(\mathcal{A})$, the canonical map

$$\operatorname{RHom}(L, M) \longrightarrow \operatorname{RHom}(\varphi L, \varphi M)$$

is an isomorphism,

as is seen at once by a spectral sequence argument.

2.2. Noetherian schemes and divisorial schemes

Notation 2.2.1. Let S be a scheme. We write $\mathrm{Qcoh}(S)$, respectively $\mathrm{Coh}(S)$, for the abelian category of quasi-coherent, respectively coherent, $\mathcal{O}_S$-modules. Let $[m,n]$ be an interval of $\mathbb{Z}$. We shall say that an object E of $\mathrm{D}(\mathrm{Qcoh}(S))$ is n-pseudo-coherent, respectively pseudo-coherent, respectively of perfect amplitude contained in $[m,n]$, respectively perfect, if it is so relative to the category fibered over the Zariski site of S defined by

$$U \longmapsto \mathrm{Loclib}(U)$$

((I 2.3), (I 4.7), and (2.0)). We denote by $\mathrm{D}^-(\mathrm{Qcoh}(S))_{\mathrm{coh}}$, respectively $\mathrm{D}(\mathrm{Qcoh}(S))_{\mathrm{parf}}$, the full triangulated subcategory of $\mathrm{D}(\mathrm{Qcoh}(S))$ formed by pseudo-coherent, respectively perfect, complexes.

Proposition 2.2.2. Let S be a noetherian scheme. The canonical functor

$$\mathrm{D}^-(\mathrm{Coh}(S)) \longrightarrow \mathrm{D}(\mathrm{Qcoh}(S))$$

is fully faithful and has as essential image $\mathrm{D}^-(\mathrm{Qcoh}(S))_{\mathrm{coh}}$.

Proof. Apply (1.1), taking for the site the Zariski site of S, and for $\mathcal{C}$, respectively $\mathcal{C}_0$, the fibered categories $U \mapsto \mathrm{Qcoh}(U)$, respectively $U \mapsto \mathrm{Coh}(U)$. Since $\mathcal{O}_S$ is coherent, the objects of $\mathcal{C}_0$ are pseudo-coherent (I 3.5); hence an object of $\mathrm{D}(\mathrm{Qcoh}(U))$ is pseudo-coherent, in the sense of (2.2.1), if and only if it is pseudo-coherent relative to $\mathcal{C}_0$ (I 2.14.1). By (1.1 c)), it remains to verify conditions (i) and (ii) of (1.1). For (ii) this is trivial, since a quasi-coherent module which is a quotient of a coherent module is coherent, and coherence is local. For (i), one uses (2.1.1 a)) and the fact that every quasi-coherent $\mathcal{O}_S$-module is a filtered inductive limit of coherent $\mathcal{O}_S$-modules (EGA I 9.4.9).

Granting (3.5) and (3.7), which are independent of the present section, one easily deduces the following.

Corollary 2.2.2.1. Let S be a noetherian scheme, respectively a noetherian scheme of finite dimension. The canonical functor

$$\mathrm{D}^b(\mathrm{Coh}(S)) \longrightarrow \mathrm{D}(S)$$

respectively

$$\mathrm{D}^-(\mathrm{Coh}(S)) \longrightarrow \mathrm{D}(S)$$

is fully faithful and has as essential image $\mathrm{D}^b(S)_{\mathrm{coh}}$, respectively $\mathrm{D}^-(S)_{\mathrm{coh}}$ (I 2.15).

Proposition 2.2.3 (cf. EGA II 4.5.2 and 4.5.5). Let S be a quasi-compact and quasi-separated scheme, and let $(L_i)_{i\in I}$ be a family of invertible sheaves on S. For every $\mathcal{O}_S$-module F and every pair $(i,n) \in I \times \mathbb{Z}$, let $F_{(i,n)}$ denote the sub-$\mathcal{O}_S$-module of $F \otimes L_i^{\otimes n}$ generated by the sections of $F \otimes L_i^{\otimes n}$ over S. The following conditions are equivalent:

(i) the open sets S_f, for $f \in \Gamma(S, L_i^{\otimes n})$, $i \in I$ and $n \in \mathbb{N}$, form a base for the topology of S;

(ii) those of the S_f which are affine cover S;

(iii) for every quasi-coherent $\mathcal{O}_S$-module F of finite type, there exists a finite family (i_α, n_α) with $n_\alpha \geq 0$ and an epimorphism

$$\bigoplus_\alpha L_{i_\alpha}^{\otimes(-n_\alpha)} \longrightarrow F;$$

(iv) for every $F \in \operatorname{ob}\operatorname{Qcoh}(S)$, one has

$$F = \sum_{(i,n)\in I\times\mathbb{N}} F_{(i,n)} \otimes L_i^{\otimes(-n)},$$

a sum, not necessarily direct, of sub-sheaves of F;

(v) same condition as (iv), but with F a quasi-coherent ideal of $\mathcal{O}_S$.

Proof. This is a simple paraphrase of the cited passages; we recall the argument briefly for the reader's convenience. For (i)⇒(ii), choose for $x \in S$ an affine open U containing x. An open S_f with $x \in S_f \subset U$, where $f \in \Gamma(S, L_i^{\otimes n})$ and $n \geq 0$, is necessarily affine, by the following lemma.

Lemma 2.2.3.1 (EGA II 5.5.8). Let X be a scheme, L an invertible $\mathcal{O}_X$-module, and $f \in \Gamma(X, L)$. The canonical immersion $X_f \to X$ is affine.

Proof. This is trivial: the question is local on X, so one may suppose X affine and $L \simeq \mathcal{O}_X$.

For (ii)⇒(iii), choose a finite covering of S by affine opens $S_{f_{ij}}$, with $f_{ij} \in \Gamma(S, L_i^{\otimes m_{ij}})$. Replacing the f_{ij} by suitable powers, one may suppose that the m_{ij} with fixed i are all equal to an integer m_i. If F is quasi-coherent of finite type, then, for each i, j, there is an integer $q_i \geq 0$ such that every section of F over $S_{f_{ij}}$ extends to a section of $F \otimes L_i^{\otimes q_i m_i}$ over S. Thus $F \otimes L_i^{\otimes q_i m_i}$ is generated over $S_{f_{ij}}$ by finitely many global sections. Proceeding as in the proof of EGA II 4.5.5, one obtains, for each i, an integer $n_i \geq 0$ such that $F \otimes L_i^{\otimes n_i}$ is generated over the union of the corresponding $S_{f_{ij}}$ by finitely many sections over S. Hence there are integers $r_i \geq 0$ and a morphism

$$L_i^{-n_i\, r_i} \longrightarrow F$$

which induces an epimorphism over that union; the assertion follows.

For (iii)⇒(iv), since every quasi-coherent sheaf on S is the inductive limit of its quasi-coherent subsheaves of finite type (EGA I 9.4.9 and IV 1.7.7), it is enough to treat the case where F is of finite type. There exist morphisms

$$u_i : L_i^{-n_i\, r_i} \longrightarrow F$$

whose sum is an epimorphism. But

$$\operatorname{Im}(u_i) = \operatorname{Im}\bigl(u_i \otimes \operatorname{Id}(L_i^{n_i})\bigr) \otimes L_i^{-n_i},$$

and $\operatorname{Im}\bigl(u_i \otimes \operatorname{Id}(L_i^{n_i})\bigr)$ is a subsheaf of $F_{(i,n_i)}$, which gives the assertion. The implication (iv)⇒(v) is trivial. Finally, for (v)⇒(i), let $x \in S$, let U be an open containing x, and let F be a quasi-coherent ideal defining $S - U$. By (v), there are $(i,n) \in I \times \mathbb{N}$ and a section f of $F \otimes L_i^{\otimes n}$ over S with $f(x) \neq 0$. Then $x \in S_f \subset U$, proving (i).

Definition 2.2.4. Let S be a quasi-compact and quasi-separated scheme, and let $(L_i)_{i\in I}$ be a family of invertible sheaves on S. We say that the L_i form an *ample family* if the equivalent conditions of (2.2.3) hold. In particular, a family consisting of a single sheaf L is ample if and only if L is ample in the usual sense (EGA II 4.5.3).

Definition 2.2.5. A scheme S is called *divisorial* if S is quasi-compact and quasi-separated and if there exists an ample family (2.2.4) of invertible $\mathcal{O}_S$-modules.

Lemma 2.2.6. Let X be a locally noetherian scheme. Let U be an open of X such that the inclusion $U \to X$ is affine, for example U affine and X separated. Then $X - U$ is purely of codimension 0 or 1; that is, at every maximal point x of $X - U$ one has $\dim \mathcal{O}_{X,x} = 0$ or 1. It is purely of codimension 1 if U contains the maximal points of X.

Proof. Put $Y = X - U$. Let x be a maximal point of Y and make the base change $X' = \operatorname{Spec} \mathcal{O}_{X,x} \to X$; one obtains a cartesian diagram

$$\begin{tikzcd} U' \arrow[r] \arrow[d] & U \arrow[d] \\ X' \arrow[r] & X. \end{tikzcd}$$

Let $Y' = X' - U'$. Since x is a maximal point of Y, Y' is just the closed point of X'. Since X' is affine, U' is an affine open of X'. We show that the dimension d of X' is 0 or 1. Indeed, one knows (SGA 2 V 5.7) that $H^d_{Y'}(X', \mathcal{O}_{X'}) \neq 0$. But if $d \geq 2$, then

$$H^d_{Y'}(X', \mathcal{O}_{X'}) = H^{d-1}(U', \mathcal{O}_{X'}),$$

and the latter group is zero because U' is affine and $d - 1 \geq 1$, a contradiction.

Remark 2.2.6.1. Without the separation hypothesis on X, (2.2.6) would be false for an affine open U of X, as shown by the example of the affine plane with the origin doubled (EGA I 5.5.11).

Proposition 2.2.7. Let S be a noetherian separated locally factorial scheme, that is, a scheme whose local rings are factorial. Then S is divisorial (2.2.5). In particular (EGA IV 21.11.1):

Corollary 2.2.7.1. Every noetherian regular separated scheme is divisorial.

Proof of 2.2.7. One may suppose that S is integral. Let $(S_i)_{i \in I}$ be a finite covering of S by nonempty affine opens. By (2.2.6), $S - S_i$ is purely of codimension 1; hence, by EGA IV 21.7.4, it is the support of a positive divisor. In other words, for each $i \in I$, there is an invertible sheaf L_i and a section $f_i \in \Gamma(S, L_i)$ such that $S_i = S_{f_i}$. The family (L_i) is then ample by (2.2.4), using (2.2.3 (ii)).

Remark 2.2.7.2. As in (2.2.6), the separation hypothesis on S is essential. Let S again be the affine plane with the origin doubled. Recall that S is obtained by gluing two copies of the affine plane $\mathbb{A}^2_k$ along the identity on the complement U of the origin O; hence there is a canonical projection

$$p : S \longrightarrow \mathbb{A}^2_k$$

which is an isomorphism over U, while $p^{-1}(O)$ consists of two points O_1 and O_2. An invertible sheaf on S is obtained by gluing two invertible sheaves on $\mathbb{A}^2_k$ by an isomorphism over U. Since every invertible sheaf on $\mathbb{A}^2_k$ is trivial, an invertible sheaf on S is defined by an element $g \in \Gamma(U, \mathcal{O}^*_U)$. But g is necessarily constant, since the restriction $\Gamma(\mathbb{A}^2_k, \mathcal{O}_{\mathbb{A}^2}) \to \Gamma(U, \mathcal{O}_U)$ is an isomorphism, the origin being of depth 2 in the plane. Thus every invertible sheaf on S is trivial. It is immediate, on the other hand, that p induces an isomorphism

$$\Gamma(\mathbb{A}^2_k, \mathcal{O}_{\mathbb{A}^2}) \xrightarrow{\sim} \Gamma(S, \mathcal{O}_S).$$

Consequently every global section of $\mathcal{O}_S$ which vanishes at O_1 also vanishes at O_2, and the opens S_f, for $f \in \Gamma(S, \mathcal{O}_S)$, do not form a base for the topology of S.

Theorem 2.2.8. Let S be a divisorial scheme (2.2.5), and let $(L_i)_{i \in I}$ be an ample family of invertible $\mathcal{O}_S$-modules. Let $L(S)$ be the strictly full additive subcategory of $\mathrm{Qcoh}(S)$ generated by the $L_i^{\otimes(-n)}$, for $(i, n) \in I \times \mathbb{N}$. Let $E \in \operatorname{ob} \mathrm{D}(\mathrm{Qcoh}(S))$.

a) The categories $L(S)$ and $\mathrm{Loclib}(S)$ are quasi-liftable in $\mathrm{Qcoh}(S)$ (I 1.2) and quasi-stable under kernels of epimorphisms (I 1.2).

b) Let $n \in \mathbb{Z}$. For E to be n-pseudo-coherent, respectively pseudo-coherent, in the sense of (2.2.1), it is necessary and sufficient that E be so relative to $L(S)$ or to $\mathrm{Loclib}(S)$ (I 2.3). The canonical functors

$$\mathrm{K}^-(L(S))/\mathrm{K}^{-,\delta}(L(S)) \longrightarrow \mathrm{D}(\mathrm{Qcoh}(S)),$$
$$\mathrm{K}^-(\mathrm{Loclib}(S))/\mathrm{K}^{-,\delta}(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}(\mathrm{Qcoh}(S))$$

are fully faithful and have as essential image $\mathrm{D}^-(\mathrm{Qcoh}(S))_{\mathrm{coh}}$.

c) Let $[m, n]$ be an interval of $\mathbb{Z}$. For E to have perfect amplitude contained in $[m, n]$ (2.2.1), it is necessary and sufficient that E be isomorphic, in $\mathrm{D}(\mathrm{Qcoh}(S))$, to a complex F such that $F^i = 0$ for $i \notin [m, n]$, $F^i \in \mathrm{ob}\, L(S)$ for $i > m$, and F^m is locally free of finite type. The canonical functor

$$\mathrm{K}^b(\mathrm{Loclib}(S))/\mathrm{K}^{b,\delta}(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}(\mathrm{Qcoh}(S))$$

is fully faithful and has as essential image $\mathrm{D}(\mathrm{Qcoh}(S))_{\mathrm{parf}}$.

Proof. Let $F \to G$ be an epimorphism in $\mathrm{Qcoh}(S)$, with G of finite type. Since F is the inductive limit of its quasi-coherent subsheaves of finite type (EGA I 9.4.9), there exists, by (2.1.1 a)), a quasi-coherent subsheaf F' of finite type in F such that $F' \to F \to G$ is an epimorphism. By (2.2.3 (iii)), there is an epimorphism $E \to F'$, with $E \in \mathrm{ob}\, L(S)$; hence $E \to F' \to F \to G$ is an epimorphism. This proves that the category of quasi-coherent $\mathcal{O}_S$-modules of finite type is quasi-liftable in $\mathrm{Qcoh}(S)$, and a fortiori the same is true for $\mathrm{Loclib}(S)$ and for $L(S)$.

For the second assertion of a), apply (1.1) on the Zariski site of S, with $\mathcal{C}$ the fibered category $U \mapsto \mathrm{Qcoh}(U)$ and $\mathcal{C}_0$ the fibered category $U \mapsto L(U)$. Since every object of L is pseudo-coherent, the notions of n-pseudo-coherence and pseudo-coherence in the fibered category $\mathrm{D}(\mathrm{Qcoh})$ coincide with the same notions relative to L (I 2.14). Condition (ii) of (1.1) is satisfied by (2.2.3 (iii)), and thus (1.1 a)) gives the part of a) concerning $L(S)$. The assertion for $\mathrm{Loclib}(S)$ is proved similarly. Part b) follows immediately from (1.1 b), c)), taking into account that, if $E \in \mathrm{ob}\, \mathrm{D}(\mathrm{Qcoh}(S))$ is n-pseudo-coherent, then E lies in $\mathrm{D}^-(\mathrm{Qcoh}(S))$, since S is quasi-compact. An object of the fibered category Qcoh is locally in L if and only if it is locally free of finite type; hence L is locally liftable in Qcoh and locally stable under kernels of epimorphisms. Thus c) follows from (1.2), with $\mathcal{C} = \mathrm{Qcoh}$, $\mathcal{C}_0 = L$, and $\mathcal{C}_1 = \mathrm{Loclib}(S)$.

Proposition 2.2.9. The hypotheses and notation are those of (2.2.8), but assume moreover that S is separated or noetherian.

a) The canonical functors

$$\mathrm{K}^-(L(S))/\mathrm{K}^{-,\delta}(L(S)) \longrightarrow \mathrm{D}(S), \qquad \mathrm{K}^-(\mathrm{Loclib}(S))/\mathrm{K}^{-,\delta}(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}(S)$$

are fully faithful and have as essential image $\mathrm{D}^-(S)_{\mathrm{coh}}$. Moreover, if S is of finite cohomological dimension in the sense of (3.7), for example if the underlying space of S is noetherian of finite Krull dimension, then the functors

$$\mathrm{K}(L(S))/\mathrm{K}^{\delta}(L(S)) \longrightarrow \mathrm{K}(\mathrm{Loclib}(S))/\mathrm{K}^{\delta}(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}^-(S)_{\mathrm{coh}}$$

are equivalences of categories.

b) The canonical functor

$$\mathrm{K}^b(\mathrm{Loclib}(S))/\mathrm{K}^{b,\delta}(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}(S)$$

is fully faithful and has as essential image $\mathrm{D}(S)_{\mathrm{parf}}$ (I 4.9). For $E \in \mathrm{ob}\, \mathrm{D}(S)$ to have perfect amplitude contained in $[m, n]$ (I 4.8), it is necessary and sufficient that E be isomorphic, in $\mathrm{D}(S)$, to a complex F such that $F^i = 0$ for $i \notin [m, n]$, $F^i \in \mathrm{ob}\, L(S)$ for $i > m$, and F^m is locally free of finite type.

Proof. In view of (3.5) and (3.7), this is an immediate corollary of (2.2.8). The only point to check, and it follows trivially from (3.5), respectively (3.7), is that for an object E of $\mathrm{D}^-(\mathrm{Qcoh}(S))$, respectively of $\mathrm{D}(\mathrm{Qcoh}(S))$ when S has finite cohomological dimension in the sense of (3.7), the notions of pseudo-coherence and perfection of (2.2.1) coincide with the usual notions (I 2.15 and 4.9) after regarding E as an object of $\mathrm{D}(S)$.

Remark 2.2.10. a) If $S = \operatorname{Spec} A$ is affine, the family consisting of the single sheaf $\mathcal{O}_S$ is ample, and for it $L(S) = \mathrm{Lib}(S)$. It is also clear that the categories $\mathrm{K}^{-,\delta}(\mathrm{Lib}(S))$ and $\mathrm{K}^{-,\delta}(\mathrm{Loclib}(S))$ are reduced to zero objects. It follows from (2.2.9) that, writing $\mathrm{Mod}(A)$ for the category of A-modules and $\mathrm{D}(A) = \mathrm{D}(\mathrm{Mod}(A))$, the functor $R\Gamma_S : \mathrm{D}(S) \to \mathrm{D}(A)$ induces fully faithful functors

$$\mathrm{D}^-(S)_{\mathrm{coh}} \longrightarrow \mathrm{D}(A), \qquad \mathrm{D}(S)_{\mathrm{parf}} \longrightarrow \mathrm{D}(A),$$

whose essential images are respectively the triangulated subcategories $\mathrm{D}^-(A)_{\mathrm{coh}}$ and $\mathrm{D}(A)_{\mathrm{parf}}$, formed by complexes with bounded-above cohomology and pseudo-coherent, respectively perfect, relative to the subcategory $\mathrm{Lib}(A)$, respectively $\mathrm{Loclib}(A)$, of free, respectively projective, modules of finite type. There are commutative diagrams of equivalences of categories

$$\begin{tikzcd}
\mathrm{K}^{-,b}(\mathrm{Lib}(S)) \arrow[r] \arrow[d, "\Gamma_S"'] & \mathrm{K}^{-,b}(\mathrm{Loclib}(S)) \arrow[r] \arrow[d, "\Gamma_S"'] & \mathrm{D}^b(S)_{\mathrm{coh}} \arrow[d, "R\Gamma_S"] \\
\mathrm{K}^{-,b}(\mathrm{Lib}(A)) \arrow[r] & \mathrm{K}^{-,b}(\mathrm{Loclib}(A)) \arrow[r] & \mathrm{D}^b(A)_{\mathrm{coh}},
\end{tikzcd}$$

$$\begin{tikzcd}
\mathrm{K}^b(\mathrm{Loclib}(S)) \arrow[r] \arrow[d, "\Gamma_S"'] & \mathrm{D}(S)_{\mathrm{parf}} \arrow[d, "R\Gamma_S"] \\
\mathrm{K}^b(\mathrm{Loclib}(A)) \arrow[r] & \mathrm{D}(A)_{\mathrm{parf}}.
\end{tikzcd}$$

In the upper diagram, one may replace b by $-$ when S has finite cohomological dimension in the sense of (3.7).

b) For an arbitrary scheme S, it follows from a) that a complex of $\mathcal{O}_S$-modules which is pseudo-coherent and has bounded cohomology is locally isomorphic, in $\mathrm{D}(S)$, to a strictly pseudo-coherent complex, which was by no means evident a priori (cf. I 2.3.1).

2.3. Soft sheaves

2.3.1. Recall a few definitions and results from [T.F.]. Let X be a paracompact space. A sheaf of sets F on X is called *soft* if, for every closed subset Y of X, every section of F over Y extends to a section of F over X. Soft abelian sheaves are acyclic: if F is soft, then $H^i(X, F) = 0$ for $i > 0$ (T.F. II 4.4.3).

Let A be a sheaf of rings on X. If F is a soft A-module, then F is generated by its sections (cf. 2.1.1): indeed, the homomorphism $A_X^{(I)} \to F$ defined by $I = H^0(X, F)$ is an epimorphism. Moreover, if A is soft, every A-module is soft (T.F. II 3.7.1). Examples of soft sheaves of rings include the sheaf of continuous real or complex-valued functions on a paracompact space and the sheaf of C^∞ functions on a C^∞ manifold.

Proposition 2.3.2. Let S be a compact space equipped with a soft sheaf of rings A. Write $\mathrm{Mod}(S)$ for the category of left A-modules and $\mathrm{D}(S) = \mathrm{D}(\mathrm{Mod}(S))$.

a) The category $\mathrm{Lib}(S)$, respectively $\mathrm{Loclib}(S)$ (2.0), is liftable in $\mathrm{Mod}(S)$ and quasi-stable, respectively stable, under kernels of epimorphisms (I 1.2). This assertion holds more generally for S paracompact.

b) Let $E \in \operatorname{ob} \mathrm{D}(S)$ and $n \in \mathbb{Z}$. For E to be n-pseudo-coherent, respectively pseudo-coherent, it is necessary and sufficient that it be so relative to $\mathrm{Lib}(S)$ or to $\mathrm{Loclib}(S)$. The canonical functors

$$\mathrm{K}^-(\mathrm{Lib}(S)) \longrightarrow \mathrm{D}(S), \qquad \mathrm{K}^-(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}(S)$$

are fully faithful and have as essential image $\mathrm{D}^-(S)_{\mathrm{coh}}$ (I 2.15).

c) Let $E \in \operatorname{ob} \mathrm{D}(S)$ and let $[m, n]$ be an interval of $\mathbb{Z}$. For E to have perfect amplitude contained in $[m, n]$ (I 4.8), it is necessary and sufficient that there exist a quasi-isomorphism $F \to E$, where F is a complex such that $F^i = 0$ for $i \notin [m, n]$, $F^i \in \operatorname{ob} \mathrm{Lib}(S)$ for $i > m$, and $F^m \in \operatorname{ob} \mathrm{Loclib}(S)$. The canonical functor

$$\mathrm{K}^b(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}(S)$$

is fully faithful and has as essential image $\mathrm{D}(S)_{\mathrm{parf}}$ (I 4.9).

d) Let $\mathrm{Mod}(A(S))$ be the category of left $A(S)$-modules and $\mathrm{D}(A(S))$ the corresponding derived category. The functor

$$H^0(S, -) : \mathrm{Mod}(S) \longrightarrow \mathrm{Mod}(A(S))$$

is exact and is an equivalence of categories. Moreover it defines equivalences of categories

$$\mathrm{D}(S) \xrightarrow{\sim} \mathrm{D}(A(S)), \qquad \mathrm{D}^-(S)_{\mathrm{coh}} \xrightarrow{\sim} \mathrm{D}^-(A(S))_{\mathrm{coh}}, \qquad \mathrm{D}(S)_{\mathrm{parf}} \xrightarrow{\sim} \mathrm{D}(A(S))_{\mathrm{parf}},$$

where pseudo-coherence, respectively perfection, in $\mathrm{D}(A(S))$ is understood relative to the category of free, respectively projective, modules of finite type. It also gives an equivalence from the full subcategory of $\mathrm{D}(S)$ formed by n-pseudo-coherent objects, respectively by objects of perfect amplitude contained in $[m, n]$, onto the analogous full subcategory of $\mathrm{D}(A(S))$.

Proof. The first assertion of a) follows from (2.1.2 a)), applied with S the topos of sheaves of sets on S and $\mathcal{C}'_S = \mathrm{Mod}(S)$, using (2.3.1). The second follows from (1.1 a)), respectively (1.2 a)), with $\mathcal{C} = \mathrm{Mod}$ and $\mathcal{C}_0 = \mathrm{Lib}$, respectively Loclib; the condition (ii) of (1.1) is verified by (2.3.1) and (2.1.1 b)). Part b) follows from (1.1 b), c)) with the same $\mathcal{C}$ and $\mathcal{C}_0$. Part c) follows from (1.2 b), c), d)), with $\mathcal{C} = \mathrm{Mod}$ and $\mathcal{C}_0 = \mathrm{Lib}$. The first assertion of d) follows, using (2.3.1), from (2.1.2 b) (ii)) with $\mathcal{C}'_S = \mathcal{C}_S = \mathrm{Mod}(S)$. The other assertions in d) follow from it, together with b) and c), and from the fact that Γ_S carries free modules, respectively direct factors of free modules, of finite type to free, respectively projective, modules of finite type.

2.4. Stein spaces

Definition 2.4.1. Let S be a locally ringed topos (I 2.15.1 (ii)), with final object also denoted S. We say that S is *Stein* if the structural sheaf $\mathcal{O}_S$ is coherent (I 3.1) and if the following properties hold:

Theorem A: every coherent $\mathcal{O}_S$-module (I 3.1) is generated by its sections;

Theorem B: for every coherent $\mathcal{O}_S$-module F, one has $H^i(S, F) = 0$ for $i > 0$.

Remark 2.4.1.1. If S has enough closed points s such that the ideal $\mathfrak{m}_s$ defined by s is coherent, then the relation $H^1(S, F) = 0$ for every coherent sheaf F implies Theorem A. Indeed, let F be a coherent $\mathcal{O}_S$-module and let s be a closed point of S. There is an exact sequence

$$0 \longrightarrow \mathfrak{m}_s F \longrightarrow F \longrightarrow F/\mathfrak{m}_s F \longrightarrow 0,$$

where $F/\mathfrak{m}_s F$ is a coherent sheaf concentrated at s. Applying $H^0(S,-)$ and writing the cohomology exact sequence, one finds that

$$H^0(S,F) \longrightarrow H^0(S,F/\mathfrak{m}_s F)$$

is an epimorphism. Since

$$H^0(S,F/\mathfrak{m}_s F) = F_s/\mathfrak{m}_s F_s,$$

where $\mathfrak{m}_s$ is the maximal ideal of $\mathcal{O}_s$, Nakayama's lemma gives the conclusion. The editor does not know whether the preceding implication is true in general when one assumes only that $\mathcal{O}_S$ is a coherent sheaf of local rings. The editor also does not know whether the relation $H^1(S,F)=0$ for every coherent sheaf F implies Theorem B.

Examples 2.4.2. A complex analytic Stein space, a compact convex subset of $\mathbb{C}^n$ equipped with the sheaf induced by the sheaf of holomorphic functions, the spectrum of a noetherian ring, and an affine rigid analytic space define Stein topoi.

Lemma 2.4.3. Let S be a Stein topos (2.4.1). Suppose that the final object S is quasi-compact (2.1.1), and let $\mathrm{Coh}(S)$ be the abelian category of coherent $\mathcal{O}_S$-modules, that is, modules of finite presentation. Then:

a) $\mathrm{Lib}(S)$, respectively $\mathrm{Loclib}(S)$, is liftable in $\mathrm{Coh}(S)$ and quasi-stable, respectively stable, under kernels of epimorphisms (I 1.2).

b) The canonical functors

$$\mathrm{K}^-(\mathrm{Lib}(S)) \longrightarrow \mathrm{K}^-(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}^-(\mathrm{Coh}(S))$$

are equivalences of categories.

c) Put $A=\Gamma(S,\mathcal{O}_S)$, and write $\mathrm{Mod}(A)$ for the category of A-modules and $\mathrm{D}(A)$ for the corresponding derived category. The functor

$$\Gamma_S : \mathrm{Coh}(S) \longrightarrow \mathrm{Mod}(A)$$

is exact, fully faithful, and has as essential image the strictly full additive subcategory of $\mathrm{Mod}(A)$ formed by A-modules of finite presentation. It induces a fully faithful functor

$$\mathrm{D}^-(\mathrm{Coh}(S)) \longrightarrow \mathrm{D}(A),$$

whose essential image is the full subcategory $\mathrm{D}^-(A)_{\mathrm{coh}}$ formed by complexes pseudo-coherent relative to the subcategory of free modules of finite type.

Here "enough closed points" means enough fiber functors (SGA 4 IV 6.4.1) of the form $i^* : S \to s$, with i_* the inclusion of a closed subtopos of S; and $\mathfrak{m}_s$ denotes the sections of $\mathcal{O}_S$ whose fiber at s belongs to the maximal ideal of $\mathcal{O}_s$.

Proof. For the first assertion of a), apply (2.1.2 a)) with $\mathcal{C}'_S = \mathrm{Coh}(S)$; the hypothesis of (2.1.2) is verified by Theorem B, since if E is coherent and G is locally free of finite type, then $\mathrm{Hom}(G,E)$ is coherent. Thus $\mathrm{Loclib}(S)$, and a fortiori $\mathrm{Lib}(S)$, is liftable in $\mathrm{Coh}(S)$. The second assertion of a) follows from (1.1 a)), respectively (1.2 a)), applied with $\mathcal{C}_0 = \mathrm{Lib}$, respectively Loclib; condition (ii) of (1.1) is verified by Theorem A and (2.1.1 b)). Part b) follows from (1.1 c)), applied successively with $\mathcal{C}_0 = \mathrm{Lib}$ and $\mathcal{C}_0 = \mathrm{Loclib}$. The first assertion of c) follows from (2.1.2 b)(i)), applied with $\mathcal{C}'_S = \mathrm{Coh}(S)$. To prove that $\Gamma_S : \mathrm{D}^-(\mathrm{Coh}(S)) \to \mathrm{D}(A)$ is fully faithful, it is enough, by b), to prove that the composite

$$\mathrm{K}^-(\mathrm{Lib}(S)) \longrightarrow \mathrm{D}^-(\mathrm{Coh}(S)) \longrightarrow \mathrm{D}(A)$$

is so. But this composite factors as

$$\mathrm{K}^{-}(\mathrm{Lib}(S)) \xrightarrow{\Gamma_S} \mathrm{K}^{-}(\mathrm{Lib}(A)) \longrightarrow \mathrm{D}(A),$$

where $\mathrm{Lib}(A)$ denotes the category of free A-modules of finite type; the first arrow is an equivalence of categories, and the second is fully faithful by (I 2.7 b)), applied with $\mathcal{C} = \mathrm{Mod}(A)$ and $\mathcal{C}_0 = \mathrm{Lib}(A)$, since $\mathrm{K}^{-,\delta}(\mathrm{Lib}(A)) = 0$. This proves the last assertion of c).

Lemma 2.4.4. Under the hypotheses of (2.4.3), the canonical functor

$$\mathrm{D}^b(\mathrm{Coh}(S)) \longrightarrow \mathrm{D}(S)$$

is fully faithful and has as essential image $\mathrm{D}^b(S)_{\mathrm{coh}}$ (I 2.15).

Proof. Apply (2.1.3) with $\mathcal{A} = \mathrm{Coh}(S)$, $\mathcal{B} = \mathrm{Mod}(S)$, and $\varphi : \mathcal{A} \to \mathcal{B}$ the inclusion functor. Since this functor is fully faithful, it is enough, by (2.1.3 a)(ii)), to check that for fixed $M \in \mathrm{ob}\,\mathrm{Coh}(S)$ and $n > 0$ the functor

$$L \longmapsto \mathrm{Ext}^n_{\mathrm{Mod}(S)}(L, M)$$

is coeffaceable on $L \in \mathrm{ob}\,\mathrm{Coh}(S)$. By Theorem A and (2.1.1 b)), every coherent $\mathcal{O}_S$-module is a quotient of a free module of finite type; and if L is free of finite type, then $\mathrm{Ext}^n_{\mathrm{Mod}(S)}(L, M) = 0$ by Theorem B. The assertion follows. Hence the functor $\mathrm{D}^b(\mathrm{Coh}(S)) \to \mathrm{D}(S)$ is fully faithful and has as essential image the full subcategory of $\mathrm{D}(S)$ formed by complexes with bounded coherent cohomology, which is precisely $\mathrm{D}^b(S)_{\mathrm{coh}}$ since $\mathcal{O}_S$ is coherent (I 3.5).

Proposition 2.4.5. With the hypotheses and notation of (2.4.3), one has:

a) The canonical functors

$$\mathrm{K}^{-,b}(\mathrm{Lib}(S)) \longrightarrow \mathrm{K}^{-,b}(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}^b(S)_{\mathrm{coh}}$$

are equivalences of categories. Moreover, the functor $R\Gamma_S : \mathrm{D}^b(S) \to \mathrm{D}^b(A)$ induces a fully faithful functor

$$R\Gamma_S : \mathrm{D}^b(S)_{\mathrm{coh}} \longrightarrow \mathrm{D}^b(A),$$

whose essential image is $\mathrm{D}^b(A)_{\mathrm{coh}}$, the triangulated subcategory of $\mathrm{D}(A)$ formed by complexes with bounded cohomology and pseudo-coherent relative to the category of free modules of finite type.

b) Let $E \in \mathrm{ob}\,\mathrm{D}(S)$, and let $[m, n]$ be an interval of $\mathbb{Z}$. If E has perfect amplitude contained in $[m, n]$ (I 4.5), then E is isomorphic to a complex F such that $F^i = 0$ for $i \notin [m, n]$, $F^i \in \mathrm{ob}\,\mathrm{Lib}(S)$ for $i > m$, and $F^m \in \mathrm{ob}\,\mathrm{Loclib}(S)$.

c) The canonical functor

$$\mathrm{K}^b(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}(S)_{\mathrm{parf}}$$

is an equivalence of categories (I 4.9). Moreover, $R\Gamma_S : \mathrm{D}^b(S) \to \mathrm{D}^b(A)$ induces a fully faithful functor

$$R\Gamma_S : \mathrm{D}(S)_{\mathrm{parf}} \longrightarrow \mathrm{D}^b(A),$$

whose essential image is $\mathrm{D}^b(A)_{\mathrm{parf}}$, the triangulated subcategory of $\mathrm{D}(A)$ formed by complexes perfect relative to the category of projective modules of finite type.

Proof. a) The functors

$$\mathrm{K}^{-,b}(\mathrm{Lib}(S)) \longrightarrow \mathrm{K}^{-,b}(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}^b(\mathrm{Coh}(S)) \longrightarrow \mathrm{D}^b(A)_{\mathrm{coh}}$$

are equivalences of categories by (2.4.3 b), c). Since the fully faithful functor $\Gamma_S : \mathrm{D}^b(\mathrm{Coh}(S)) \to \mathrm{D}(A)$ factors as

$$\mathrm{D}^b(\mathrm{Coh}(S)) \longrightarrow \mathrm{D}^b(S)_{\mathrm{coh}} \xrightarrow{R\Gamma_S} \mathrm{D}(A),$$

the conclusion follows from (2.4.4).

b) Suppose $\mathrm{parf.amp}(E) \subset [m,n]$. In particular E is pseudo-coherent; by a), after replacing E by an isomorphic object, we may assume that $E^i \in \mathrm{ob}\,\mathrm{Lib}(S)$ for $i > m$ and $E^i = 0$ for $i > n$. Truncating, we may also suppose that $E^i = 0$ for $i < m$. By (I 4.16), applied with $\mathcal{C} = \mathrm{Mod}$ and $\mathcal{C}_0 = \mathrm{Lib}$, one then has $E^m \in \mathrm{ob}\,\mathrm{Loclib}(S)$.

c) The hypotheses (1.1 (i)) and, respectively, (1.1 (ii)) are verified for $\mathcal{C} = \mathrm{Coh}(S)$ and $\mathcal{C}_0 = \mathrm{Lib}$, by (2.4.3 a)) and by (2.1.1 b)) together with Theorem A. Hence (1.2 c), d)) shows that the functor

$$\mathrm{K}^b(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}^b(\mathrm{Coh}(S))$$

is fully faithful. Therefore, by (2.4.4), the same is true for

$$\mathrm{K}^b(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}(S),$$

and its essential image is $\mathrm{D}(S)_{\mathrm{parf}}$ by b). By an argument analogous to that used in the proof of (2.4.3 c)), one sees that the functor $\mathrm{K}^b(\mathrm{Loclib}(S)) \to \mathrm{D}(S)$ induced by Γ_S is fully faithful and has essential image $\mathrm{D}^b(A)_{\mathrm{parf}}$; one finishes the proof as in a).

Remark 2.4.6. The editor does not know whether in (2.4.4) one may replace D^b by D^-, and consequently whether (2.4.5 a)) remains true when one replaces $\mathrm{K}^{-,b}$ by K^- and D^b by D^-.

We finish with an interesting complement to (2.4.5). First we give a definition.

Definition 2.4.7. A *complex pseudo-analytic space* is a ringed space locally isomorphic to a ringed space of the form $(Y, \mathcal{O}_X|Y)$, where X is a complex analytic space and Y is a closed subset of X.

The structural sheaf of a complex pseudo-analytic space is therefore coherent. We say that a complex pseudo-analytic space is Stein if the corresponding topos is Stein (2.4.1).